



SmartQR: Using AI to make medical diagnosis accessible to the Indian population

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GOOGLE CLOUD RESULTS

- ✓ 30% reduction in costs
- ✓ One million+ users with further growth anticipated
- ✓ 30% improvement in operational efficiency
- ✓ 20% reduction in development time
- ✓ Accelerates time to market with AI-powered diagnostic solutions

With Google Cloud, SmartQR is delivering AI-powered mobile services at scale, making medical diagnosis more accessible to the population in India.

The healthcare sector is under increasing pressure from an aging population, skills shortages, cost and budget pressures, and administrative complexities. Fortunately, AI and cloud technologies help relieve the strain and can chart a new way forward for healthcare in the digital age.

In India, [SmartQR](#) uses these technologies to transform traditional health diagnostics and monitoring, and has turned to [Google Cloud](#) to deliver its products to a national and global market. Founded in 2021 and incubated within the Indian Council of Medical Research, the business aims to provide smartphone-based point-of-care screening tools, delivered through Android and iOS mobile applications, to detect health issues early and enable timely intervention. Used by healthcare professionals and personal users, these tools also deliver data and insights to support planning and decision-making, reduce the cost of healthcare, and minimize errors.

SmartQR has grown to around 100 employees, including specialists in AI, machine learning, medical imaging, software development, and healthcare management. The business currently offers four solutions, one each for vaginal PH, hemoglobin, or plasma levels, and one for urine parameters. Each solution comprises a detection mobile application, a dashboard for monitoring and reporting, and test strips. As of mid-2024, another six solutions were in development, all built on Google Cloud.



Evaluating cloud providers for scalability, security, and compliance

“Managing these large volumes of sensitive and diverse data, including medical images, patient records, and diagnostic reports, in accordance with strict compliance requirements, was essential to our business. This led us to Google Cloud.”

Sambit Ghosh
Founder and Group
CEO, SmartQR

To deliver its solutions, SmartQR needed a cloud provider that offers the compute capabilities and scalability to process large medical diagnostics datasets and train AI models on them. Operating in the highly regulated healthcare space, SmartQR also needed to ensure its management of patient data would comply with the European Union’s General Data Protection Regulation and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. This required robust identity-based access control, audit logs, and encryption.

“Managing these large volumes of sensitive and diverse data, including medical images, patient records, and diagnostic reports, in accordance with strict compliance requirements, was essential to our business,” says Sambit Ghosh, Founder and Group CEO.

“We also needed to integrate seamlessly with healthcare institutions, research providers, and our other technology partners so we could securely share data with them. This led us to Google Cloud.”

The Google Cloud team briefed SmartQR on relevant AI and machine learning products and high-end GPUs designed to handle the large workloads and datasets needed for its AI-powered solutions. This would accelerate model training and enhance the accuracy and efficiency of diagnoses using SmartQR products. Together, the teams determined that Google Cloud would support SmartQR's needs, enabling the business to develop and test its AI algorithm, remain within budget, and meet requirements for performance, reliability, and compliance.

To complement the expertise of Google Cloud, SmartQR selected the Google Cloud Partner [Sela](#) to assist with the deployment. "Sela has played an exceptional role across our Google Cloud journey to date, in particular getting the drivers in place for a smooth deployment," says Ghosh. "The Sela team has also supported strategies to improve infrastructure, unlock growth opportunities, and maximize cost-effectiveness."



Using Google Cloud AI to streamline medical diagnostics

“With its robust infrastructure and its pre-built AI and machine learning frameworks, Google Cloud has helped

The organization now relies on a range of Google Cloud services to provide the scalability and reliability to achieve its business objectives, delivered with usage-based billing to manage budgets. [Cloud Run](#) and [Cloud Functions](#) streamline development and minimize the need for its developers to manage infrastructure, while

**reduce our
development
time by about
20%.”**

Sambit Ghosh
Founder and Group
CEO, SmartQR

[Vertex AI](#) enables them to generate the AI and machine learning outputs that run in SmartQR’s products. BigQuery provides a robust, scalable data warehouse, with [Pub/Sub](#) providing the associated data pipelines. The organization is currently experimenting with [BigQuery ML](#) for data analytics queries.

Meanwhile, [Google Kubernetes Engine](#) enables SmartQR to run its enterprise workloads in a containerized environment at scale. For virtual machines and effective, accessible, and long-term storage, the team relies on [Compute Engine](#) and [Cloud Storage](#). To keep security and compliance front and center, the team uses [Security Command Center](#) to gain insights into the SmartQR architecture and compliance factors that improve the company’s security posture in accordance with HIPAA requirements.

Using these Google Cloud solutions with a [TensorFlow](#) machine learning platform enables SmartQR to realize its vision of delivering transformative, AI-powered healthcare services. “With its robust infrastructure and pre-built AI and machine learning frameworks, Google Cloud has helped reduce our development time by about 20%,” says Ghosh. “This has also lowered costs, enabled us to deploy multiple new features, such as colorimetry to help determine hemoglobin levels in the blood, and accelerated our go-to-market.”



Zero-compromise, scalable solutions

Google Cloud also provides the scalability SmartQR needs to manage demand spikes, which can lead to more than a million users using its AI-powered applications at one time. “Over the past three months, we have seen a tremendous response from the market, and the scalability of

Google Cloud has enabled us to handle increased user traffic and growing datasets without compromising performance and availability,” says Ghosh.

Meanwhile, automation through [Cloud Scheduler](#) and other Google Cloud services has enabled SmartQR to streamline its operational workflows, including data processing and model training and deployment. “Thanks to Google Cloud, we can be very frugal with our technical resources, and enhance operational efficiency by at least 30% compared to our previous environment,” explained Ghosh.

The business has also lowered costs by 30% thanks to resource optimization, reductions in infrastructure costs, the use of cost management tools, and support from Google Cloud.



Making medical diagnoses accessible

“Working with Google Cloud and Sela has been a transformational experience for our business. It has enabled us to develop plans to move from a

The benefits of SmartQR’s use of Google Cloud flow through to the end users of the company’s medical diagnostic products. “Traditionally, even if you needed a very basic diagnostic test, you needed to call a service provider who would come to your home, collect blood, and send you the results in up to 12 hours,” says Ghosh. “This type of service was not really accessible to many people in

preventative
healthcare
provider to a
complete
healthcare
ecosystem
provider across
global markets.”

Sambit Ghosh

Founder and Group
CEO, SmartQR

India for reasons such as
cost.”

With SmartQR’s advanced AI
services, a person just needs
an internet-connected device
to diagnose their basic
physical parameters in as little
as two minutes from the
comfort of home or at a
pharmacy, at a very low cost. If
there is a problem outside
those parameters, they can
then seek secondary or
tertiary care.

“Working with Google Cloud and Sela has been a transformational
experience for our business,” Ghosh says. “It has enabled us to
develop plans to move from a preventative healthcare provider to a
complete healthcare ecosystem provider across global markets.”

SmartQR is a pioneering force in medical AI, revolutionizing
traditional health diagnostics with its innovative AI-powered
technology. By transforming complex health data into actionable
insights, SmartQR empowers individuals to take control of their
well-being through accessible and user-friendly solutions.
Committed to democratizing healthcare, the company aims to
bridge the gap between advanced diagnostics and everyday life,
ensuring that everyone can benefit from timely and accurate
health information.

Industry: Healthcare & Life Sciences

Location: India

Products: [Google Cloud](#), [BigQuery](#), [Cloud Functions](#), [Cloud Run](#),
[Cloud Scheduler](#), [Cloud Storage](#), [Compute Engine](#), [Google](#)
[Kubernetes Engine](#), [Pub/Sub](#), [Security Command Center](#), [Vertex](#)
[AI](#)

About Google Cloud Partner- Sela

A Google Cloud Partner, Sela provides cloud managed services
and 360 degree support to business customers.

Why Google	Products and pricing	Solutions	Resources	Engage
Choosing Google Cloud	Google Cloud pricing	Infrastructure modernization	Google Cloud Affiliate Program	Contact sales
Trust and security	Google Workspace pricing	Databases	Google Cloud documentation	Find a Partner
Modern Infrastructure Cloud	See all products	Application modernization	Google Cloud quickstarts	Become a Partner
Multicloud		Smart analytics	Google Cloud Marketplace	Events
Global infrastructure		Artificial Intelligence	Learn about cloud computing	Podcasts
Customers and case studies		Security	Support	Developer Center
Analyst reports		Productivity & work transformation	Code samples	Press Corner
Whitepapers		Industry solutions	Cloud Architecture Center	Google Cloud on YouTube
Blog		DevOps solutions	Training	Google Cloud Tech on YouTube
		Small business solutions	Certifications	Follow on X
		See all solutions	Google for Developers	Join User Research
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